

19 Answer: D

$$R = \rho L/A = \rho L/(\pi r^2) = \rho(8)/(\pi(0.00025)^2) = 128 \times 10^6 \rho/\pi$$

$$R_2 = \rho(2)/(\pi(0.0005)^2) = 128 \times 10^6 \rho/\pi$$

$$\rightarrow R_2 = R$$